



Nawatech

Beyond Engineering

nawatech.co



KAI Workshop – Day 5

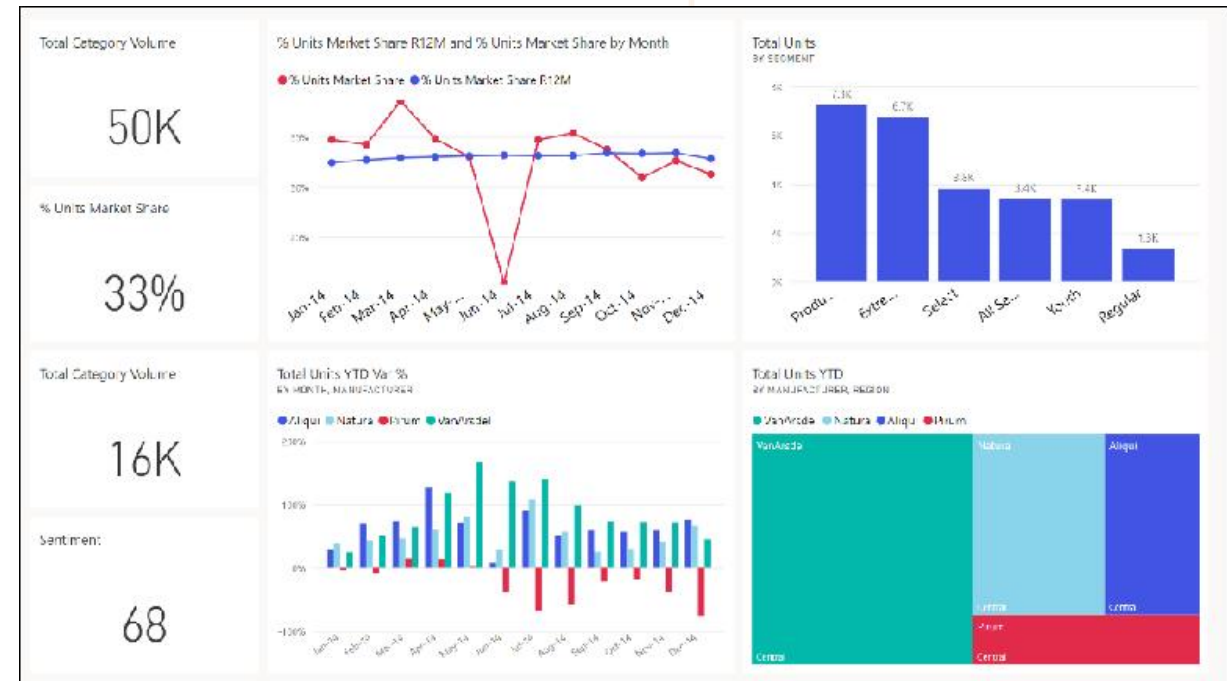
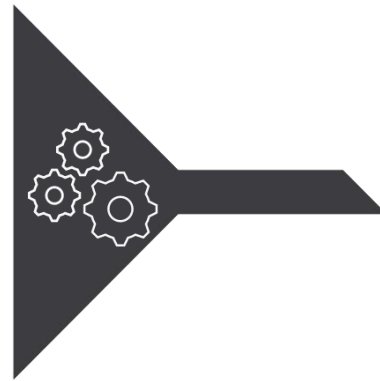
Prepared for: PT Kereta Api
Indonesia



Discover data analysis



From data to business insights with Power BI



From data to business insights with Power BI



Microsoft | Customer Stories All stories Stories by product Contact Sales All Microsoft Search

September 17, 2019

Dutch Railways transforms transport with data-driven innovations

Share the story



A new culture of innovation

With all of the customer data centrally accessible and able to be visualized using the Microsoft Power BI suite of business analytics tools, the team at Dutch Railways is now innovating like never before, coming up with new ideas for how to improve its service for customers.

"By using tools such as Microsoft Power BI, suddenly we are able to be much more agile as an organization, bringing together all sorts of different elements, insights, and data to drive innovations," says Geert. "It has shifted the attitude we have toward the way we work."

He has established an internal innovation community to bring people together from different units of the organization to discuss and develop ideas around themes like artificial intelligence, blockchain, and security.

"We start with a proof of concept and see if we can develop it into something scalable," say Geert. "That way we can see if an idea is worth investing more money into."

An extra 20,000 seats every day

One such innovation was the Seat Finder app, which uses color coding to show passengers where to stand on the platform so they can find less crowded carriages when boarding.

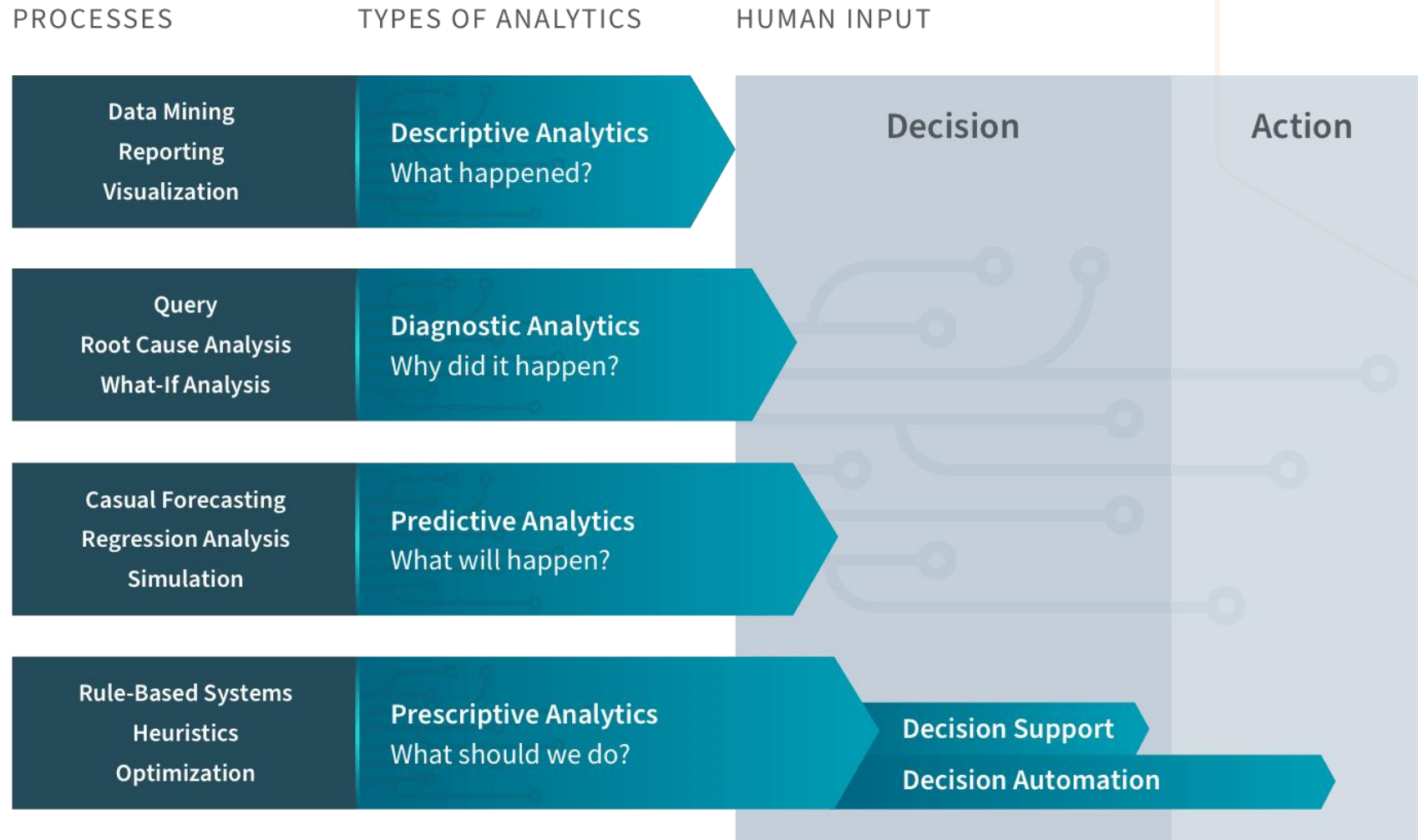


Different types of analysis

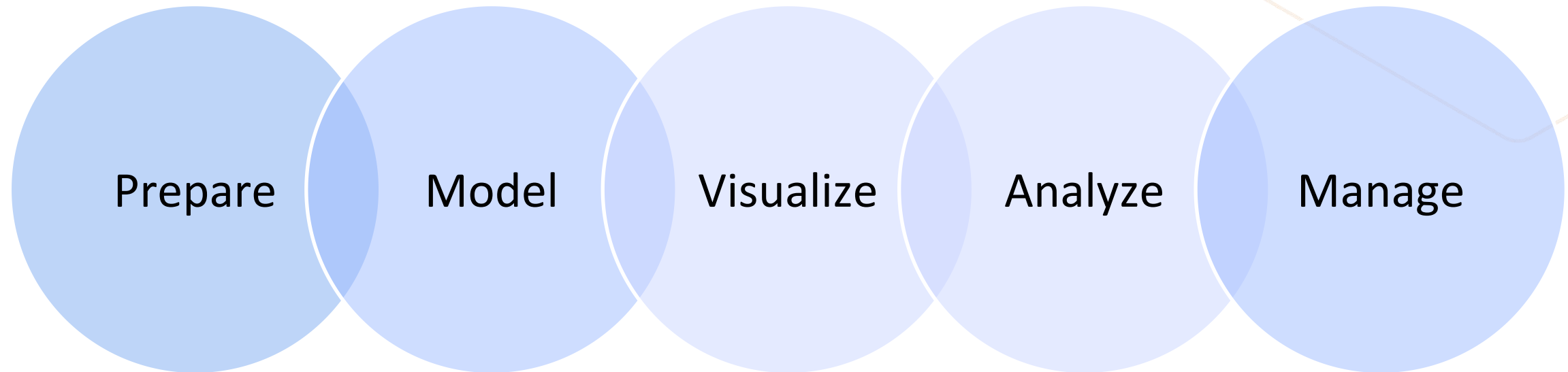
Data analysis is another form of storytelling with five categories:

- **Descriptive** : Summarize past data
- **Diagnostic** : Explain past data
- **Predictive** : Forecast future data
- **Prescriptive** : Optimize future data
- **Cognitive** : Learn from data

Different types of analysis



Tasks of a Data Analyst



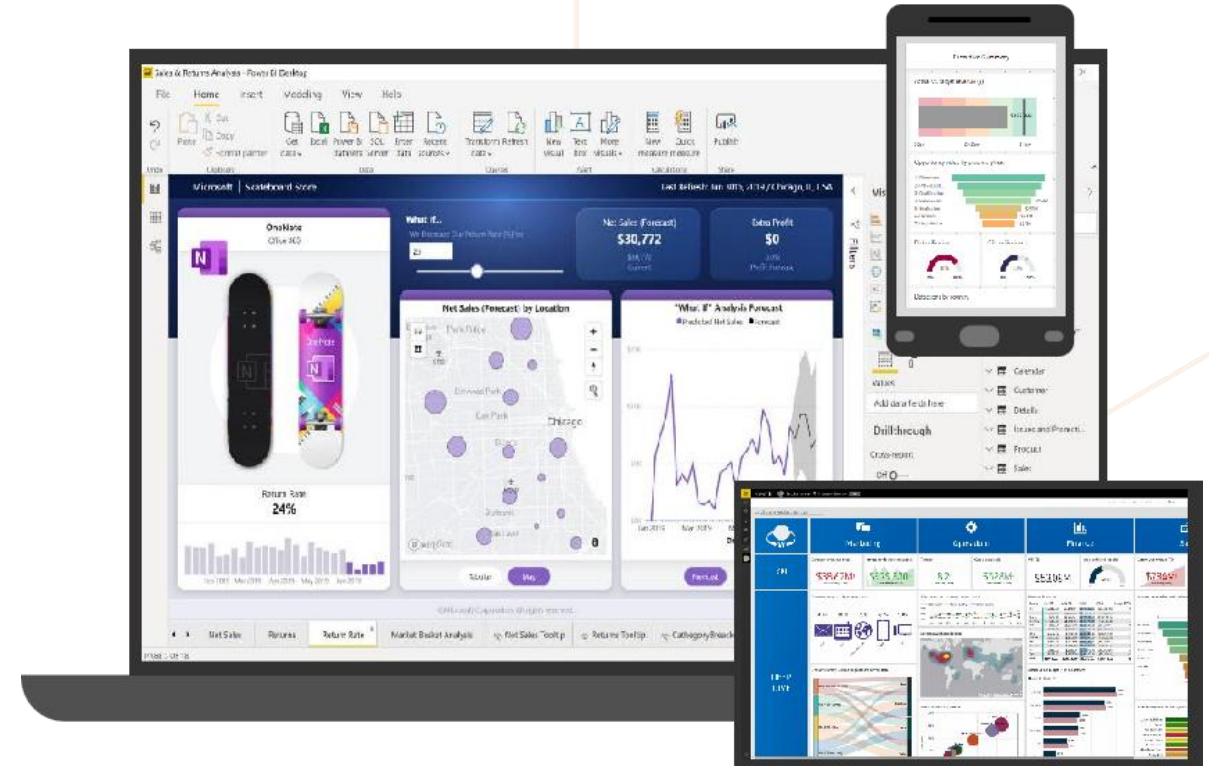
Get started building with Power BI



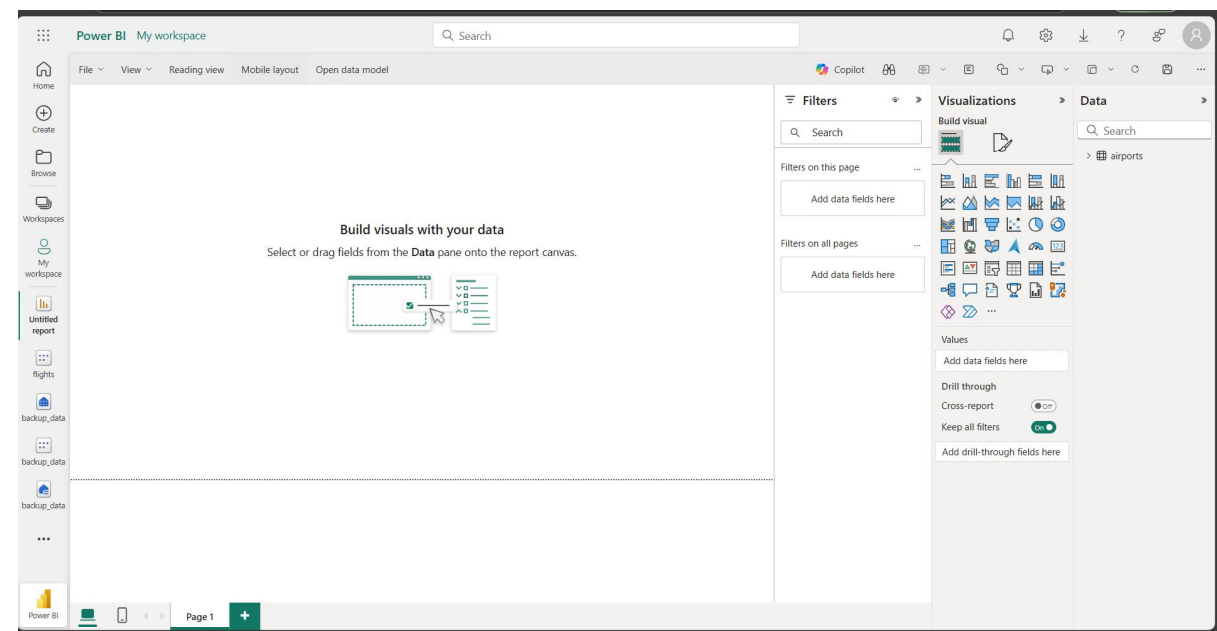
Introduction to Power BI environment



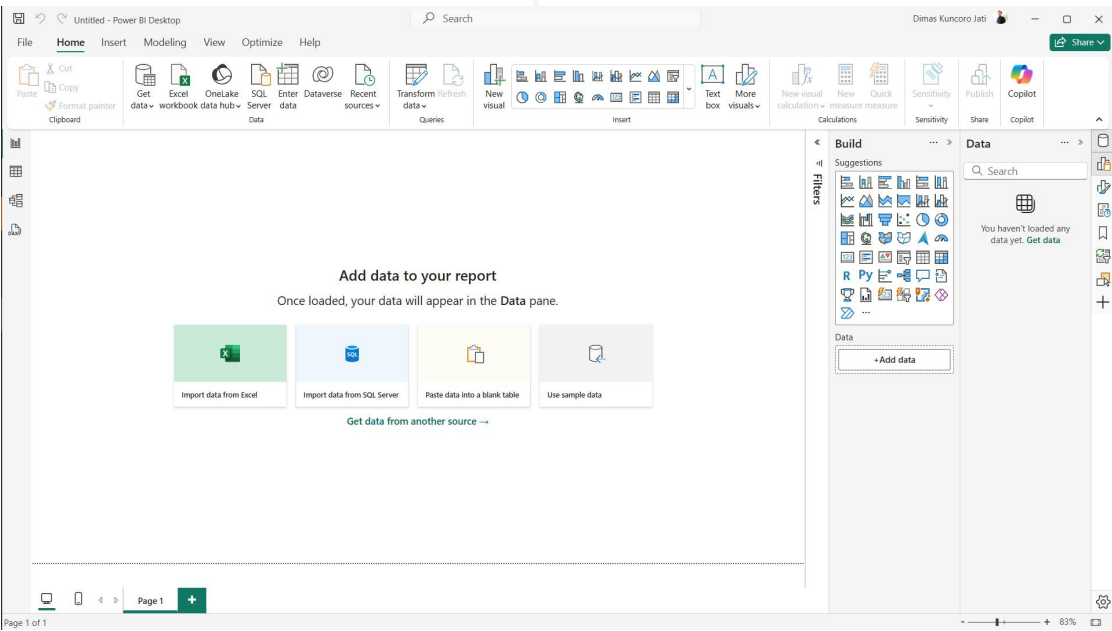
- Create with Power BI Desktop
- Distribute with Power BI service
- Access with Power BI Mobile



Power BI



Fabric



Power BI Desktop

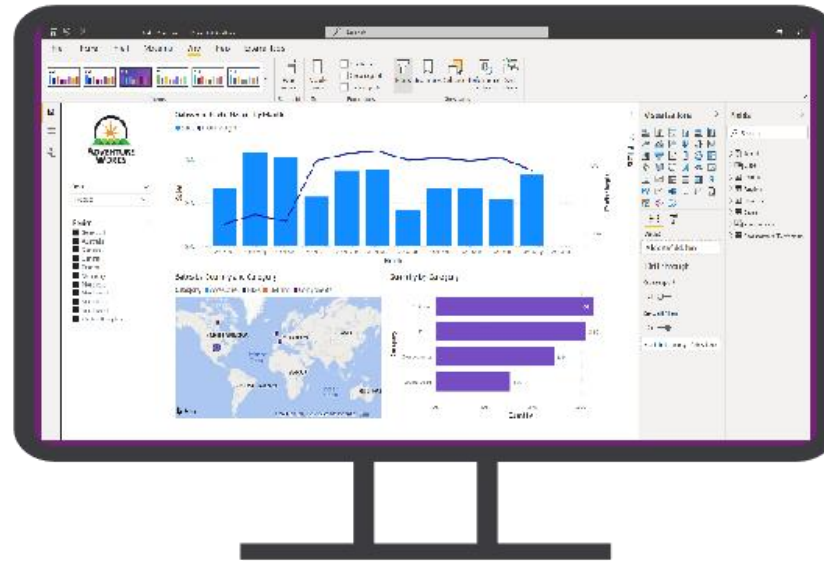
The flow of Power BI



Data sources



Power BI



Power BI service & Mobile





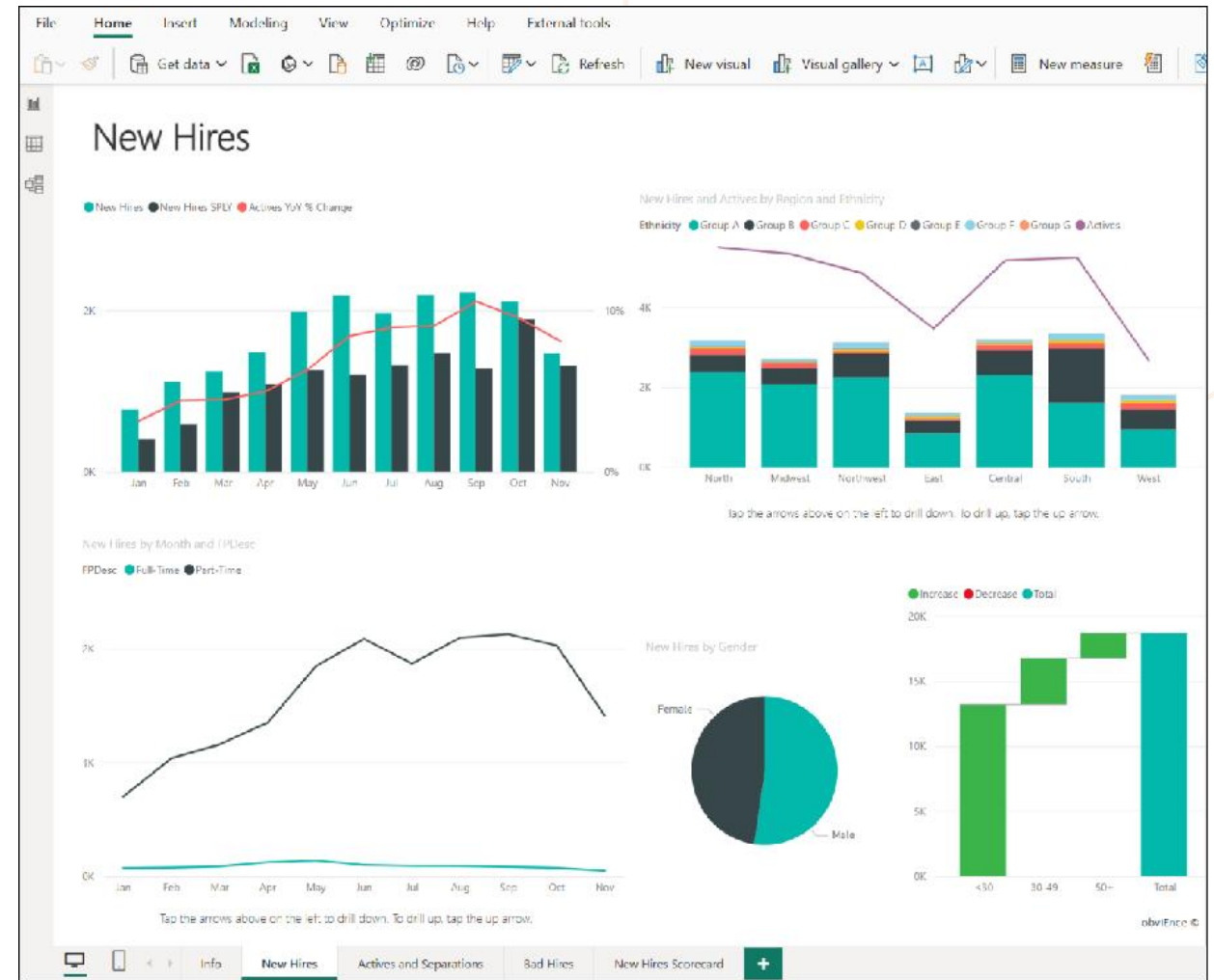
The building blocks of Power BI

Semantic models

- Connected data, transformations, relationships, and calculations.

Visualizations

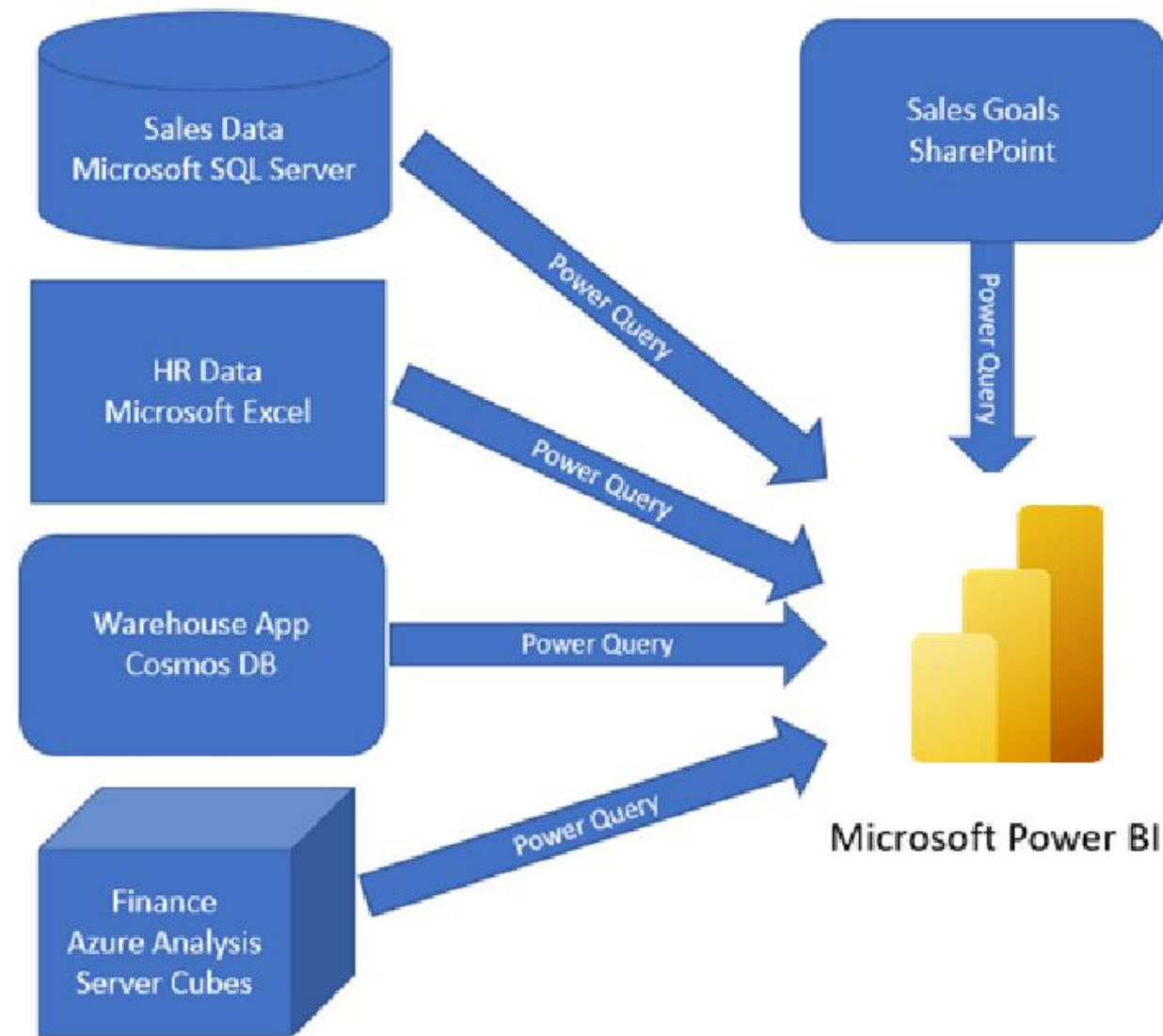
- Interactive representation of data through various charts and graphics.



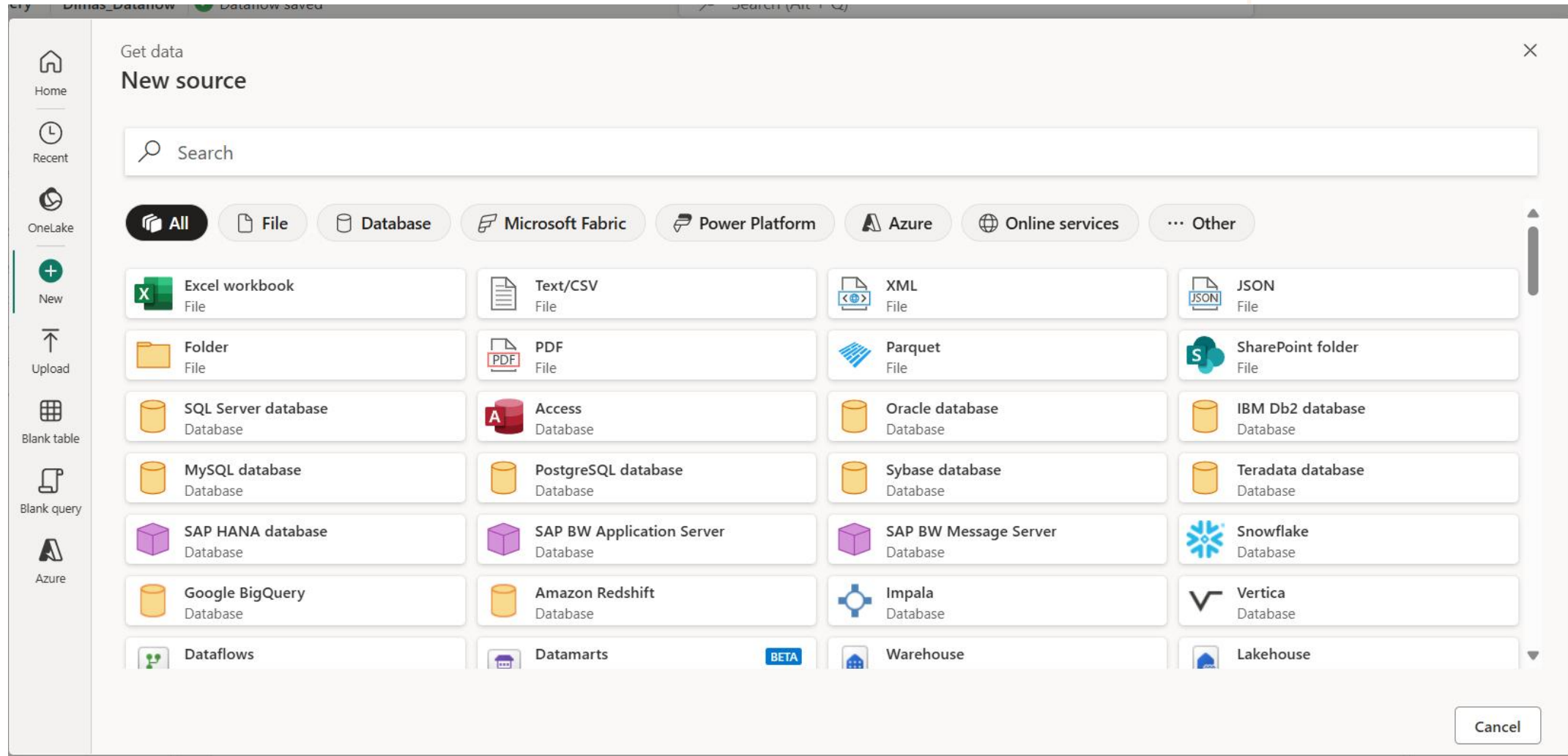
Get data in Power BI



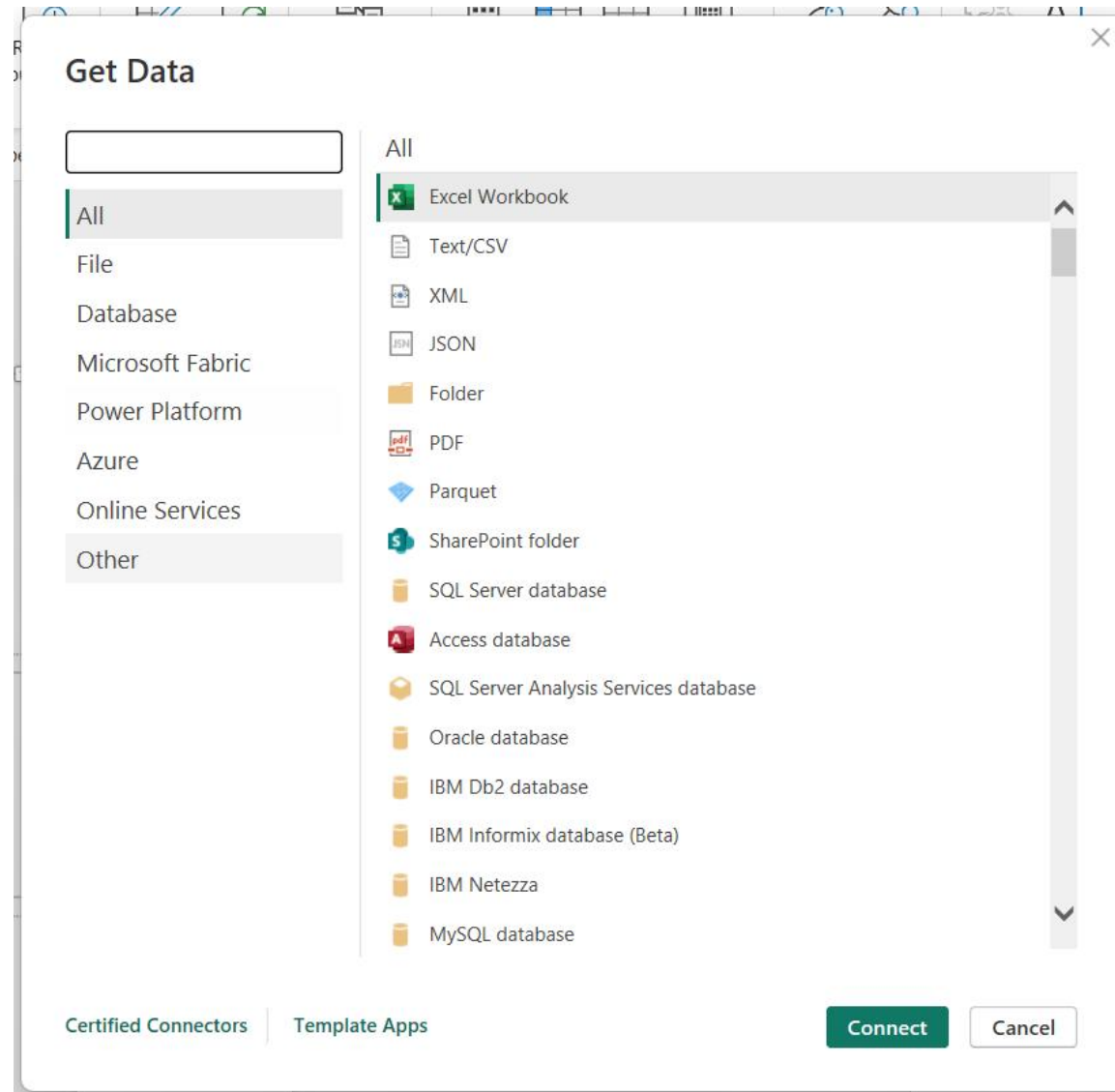
Combine all data into a single semantic model



Combine all data into a single semantic model in fabric

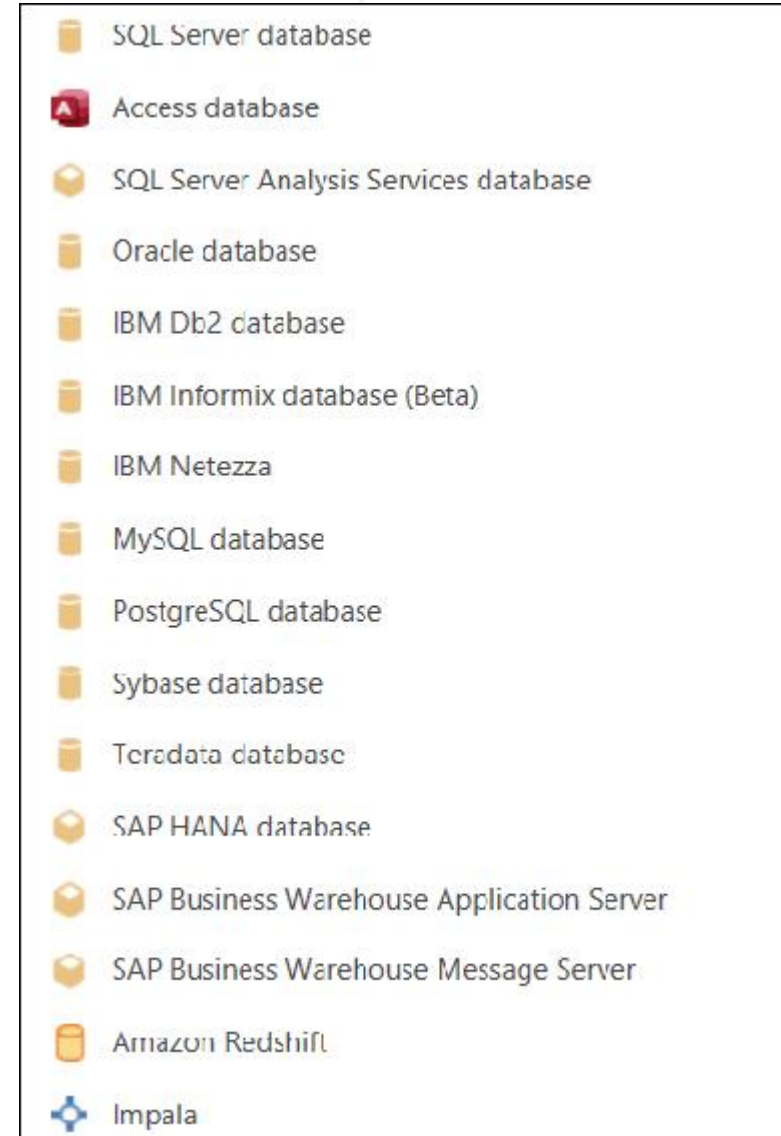
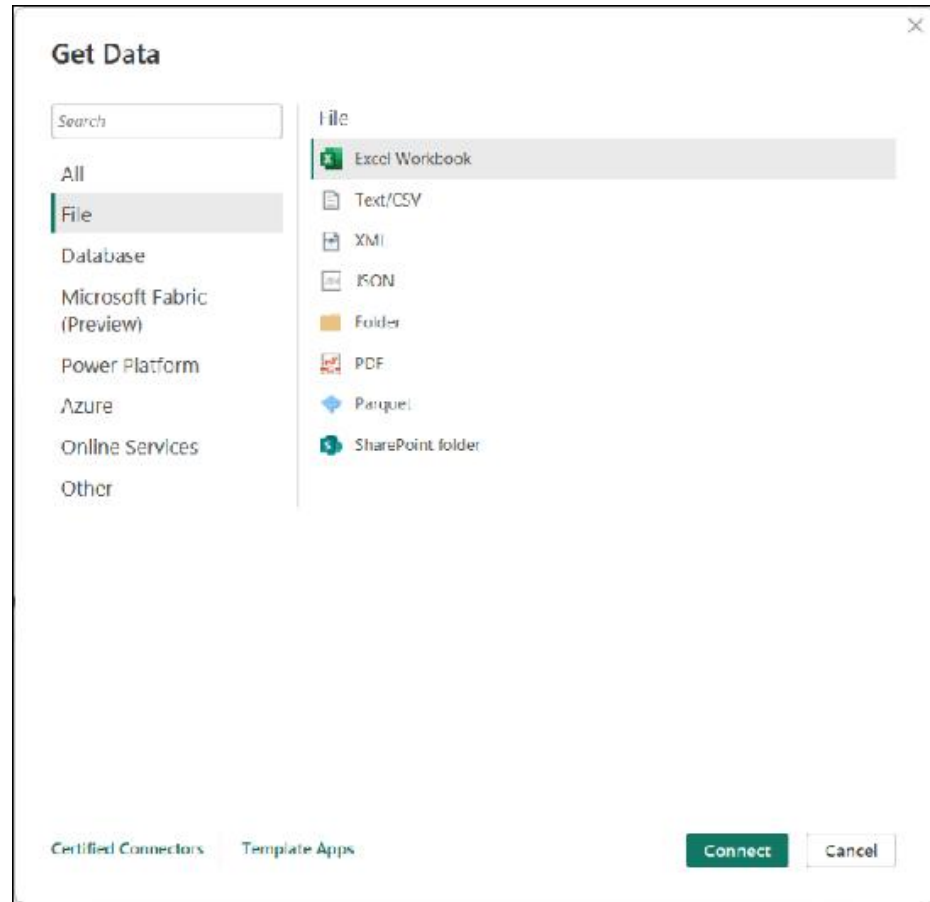


Combine all data into a single semantic model in fabric





Get data from a wide variety of sources





Select a storage mode

Storage mode affects

- Available transformations
- Report performance

*Not all sources support all modes

A screenshot of a 'Storage mode' dropdown menu. The menu is open, showing four options: 'Import', 'DirectQuery', 'Dual', and 'DirectLake'. The 'Import' option is currently selected and highlighted in a light gray background. The dropdown is contained within a light gray box with a thin black border.

Storage mode

Import

Import

DirectQuery

Dual

DirectLake

Introduction to DirectQuery



The screenshot shows the Power BI Desktop interface. The ribbon is set to the 'Home' tab. In the 'Data' group, the 'SQL Server' icon is highlighted with a red rectangle. Below the ribbon, the 'SQL Server database' dialog box is open. It contains the following fields and options:

- Server**: A text input field.
- Database (optional)**: A text input field.
- Data Connectivity mode**: Two radio button options: 'Import' and 'DirectQuery'. The 'DirectQuery' option is selected and highlighted with a red rectangle.
- Advanced options**: A collapsed section indicated by a right-pointing arrow.
- Buttons**: 'OK' and 'Cancel' buttons at the bottom right.



Implications of using DirectQuery

Benefits

- Frequently changing data
- Need near real-time
- Large data volumes
- Multi-dimensional data

Limitations

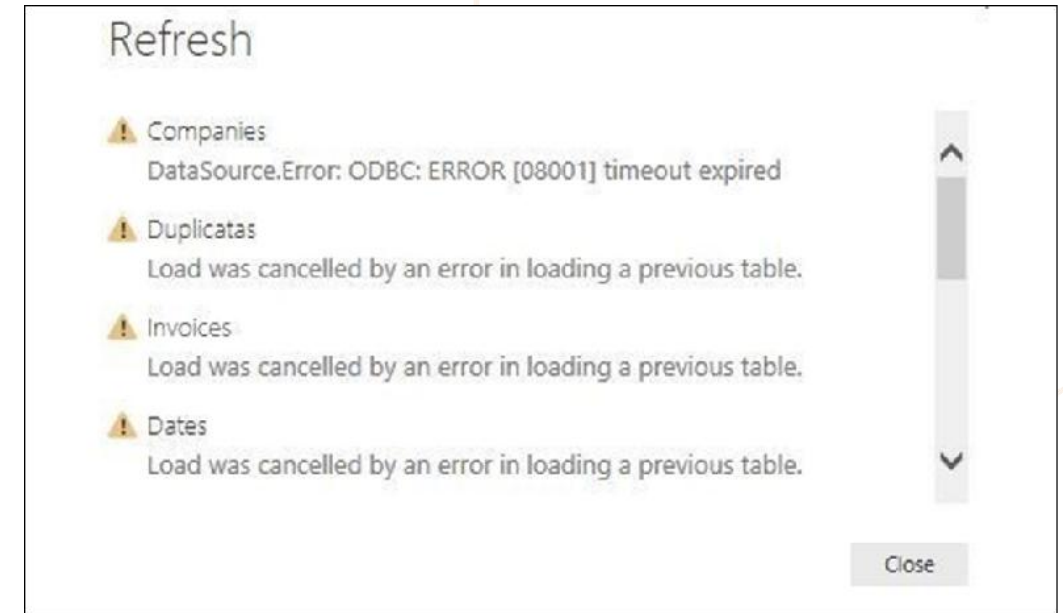
- Dependent on data source performance
- Security between source and destination
- Limited modeling capabilities
- Limited transformation features



Data import errors

Possible data load errors:

- Query Timeout
- Couldn't find data formatted as a table
- Could not find file
- Data type errors



Clean, transform, and load data



Transform data with Power Query Editor



Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Properties Advanced Editor Manage Query Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Text Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Text Analytics Vision Azure Machine Learning AI Insights

Queries [2]

- ColorFormats
- ResellerSales_202006

fx = Table.TransformColumnTypes(#"Promoted Headers",{{"SalesOrderNumber", type text}, {"SalesOrderLineNumber", Int64.Type}, {"OrderDate", type date}, {"DueDate", type date}, {"ShipDate", type date}, {"ProductKey", Int64.Type}, {"ReorderPoint", Int64.Type}})

	SalesOrderNumber	SalesOrderLineNumber	OrderDate	DueDate	ShipDate	ProductKey	ReorderPoint
1	SO71691	2	6/1/2020	6/11/2020	6/8/2020	434	
2	SO71691	4	6/1/2020	6/11/2020	6/8/2020	222	
3	SO71774	1	6/1/2020	6/11/2020	6/8/2020	436	
4	SO71774	2	6/1/2020	6/11/2020	6/8/2020	418	
5	SO71775	1	6/1/2020	6/11/2020	6/8/2020	573	
6	SO71775	2	6/1/2020	6/11/2020	6/8/2020	555	
7	SO71775	3	6/1/2020	6/11/2020	6/8/2020	490	
8	SO71776	1	6/2/2020	6/12/2020	6/9/2020	514	
9	SO71777	1	6/2/2020	6/12/2020	6/9/2020	408	
10	SO71777	2	6/2/2020	6/12/2020	6/9/2020	436	
11	SO71778	1	6/2/2020	6/12/2020	6/9/2020	467	
12	SO71778	2	6/2/2020	6/12/2020	6/9/2020	566	
13	SO71778	3	6/2/2020	6/12/2020	6/9/2020	554	
14	SO71778	4	6/2/2020	6/12/2020	6/9/2020	497	
15	SO71779	1	6/2/2020	6/12/2020	6/9/2020	589	
16	SO71779	2	6/2/2020	6/12/2020	6/9/2020	543	
17							

14 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 12:52 PM

Query Settings

PROPERTIES

Name
ResellerSales_202006

All Properties

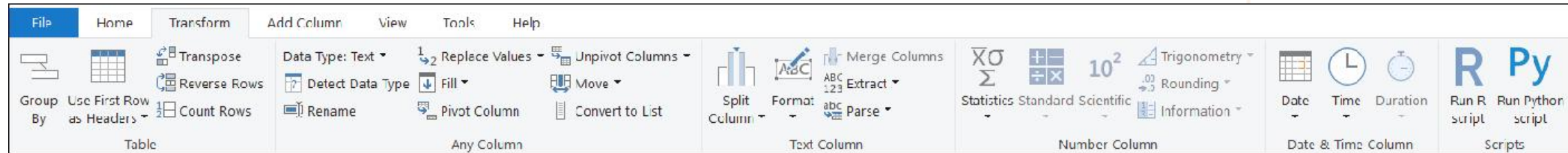
APPLIED STEPS

- Source
- Promoted Headers
- Changed Type

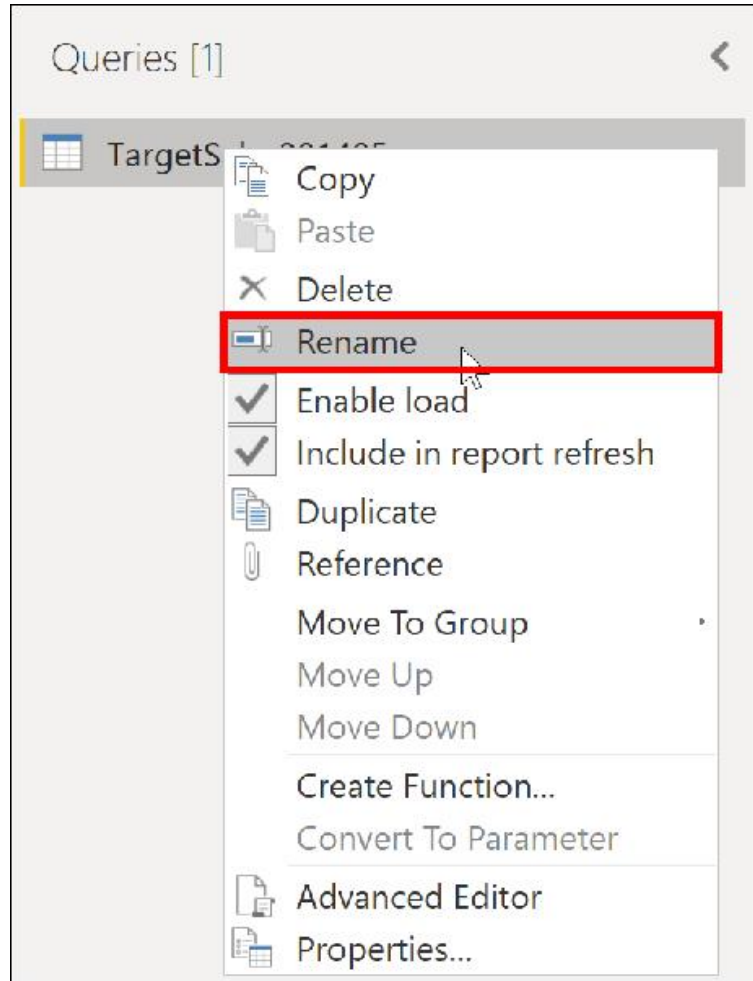


Common transformations

- Transform columns, add new, split, extract, and more



Choose user-friendly values



The screenshot shows the 'Transform' ribbon in a data tool. The 'Replace Values' button is highlighted with a red rectangle. Below the ribbon is a data table with columns 'ID', 'Subcategory Name', and 'Attribute'.

	ID	Subcategory Name	Attribute
1	1	Mountain Bikes	January
2	1	Mountain Bikes	February
3	1	Mountain Bikes	March
4	1	Mountain Bikes	April
5	1	Mountain Bikes	May
6	1	Mountain Bikes	June
7	1	Mountain Bikes	July
8	1	Mountain Bikes	August
9	1	Mountain Bikes	September
10	1	Mountain Bikes	October
11	1	Mountain Bikes	November
12	1	Mountain Bikes	Dezember
13	2	Road Bikes	January
14	2	Road Bikes	February

Shaping table structure



Advanced Editor

Manage ▾

Query

Choose Columns ▾ Remove Columns ▾

Keep Rows ▾ Remove Rows ▾

Reduce Rows ▾ Sort ▾

Remove Columns

Remove Other Columns

	A ^B _C Column13	A ^B _C Column14
	November	December
	880000	890000
	9500	10000
	511000	512000

Properties

Advanced Editor

Manage ▾

Query

Choose Columns ▾ Remove Columns ▾

Keep Rows ▾ Remove Rows ▾

Remove Top Rows

Remove Bottom Rows

Remove Alternate Rows

Remove Duplicates

Remove Blank Rows

Remove Errors

	A ^B _C Column13	A ^B _C Column14
	November	December
	880000	890000
	9500	10000
	511000	512000

Evaluate and change column data types



ABC SalesOrderNumber	ABC OrderDate	ABC TotalCost
SO43659	7/1/2017	2024.99
SO43659	7/1/2017	6074.97
SO43659	7/1/2017	2024.99
SO43659	7/1/2017	2039.99
SO43659	7/1/2017	2039.99
SO43659	7/1/2017	4079.98
SO43659	7/1/2017	2039.99
SO43659	7/1/2017	86.52
SO43659	7/1/2017	28.84
SO43659	7/1/2017	34.2

Couldn't load the data for this visual

MdxScript(Model) (19, 40) Calculation error in measure

'Sales'[Quantity of Orders YTD]: A column specified in the call to function 'TOTALYTD' is not of type DATE. This is not supported.

[Copy details](#)

Send a Frown

Close

Combine multiple queries into one



Append

Append

Concatenate rows from three or more tables into a single table.

☐ Two tables ☒ Three or more tables

Available tables

Production Suppliers

Sales Customers

HR Employees

Add >>

Tables to append

Production Suppliers

Sales Customers

HR Employees

OK

Merge

Merge

Select a table and matching columns to create a merged table.

Sales Orders

orderid	custid	empid	orderdate	requireddate	shippeddate	shipperid	freight	shipname
10248	85	5	7/4/2014	8/1/2014	7/16/2014	3	32.38	Ship to 85-B
10249	79	6	7/5/2014	8/16/2014	7/10/2014	1	11.61	Ship to 79-C
10250	34	4	7/8/2014	8/5/2014	7/12/2014	2	65.83	Destination SCC
10251	84	3	7/8/2014	8/5/2014	7/15/2014	1	41.34	Ship to B4-A

< >

Sales OrderDetails

orderid	productid	unitprice	qty	discount
10248	11	14.00	12	0
10248	42	9.80	10	0
10248	72	34.80	5	0
10249	14	18.60	9	0
10249	51	42.40	40	0

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

> Fuzzy matching options

✓ The selection matches 830 of 830 rows from the first table.

OK Cancel



Unpivot or pivot columns

Add or remove table structure to meet your aggregation needs.

	Category Name	Subcategory Name
1	Bikes	Mountain Bikes
2	Bikes	Road Bikes
3	Bikes	Touring Bikes
4	Clothing	Bib-Shorts
5	Clothing	Caps
6	Clothing	Gloves
7	Clothing	Jerseys
8	Clothing	Shorts
9	Clothing	Socks
10	Clothing	Tights
11	Clothing	Vests
12	Accessories	Bike Racks
13	Accessories	Bike Stands
14	Accessories	Bottles and Cages

	1.2 Bikes	1.2 Components	1.2 Clothing	1.2 Accessories
1	3	14	8	12

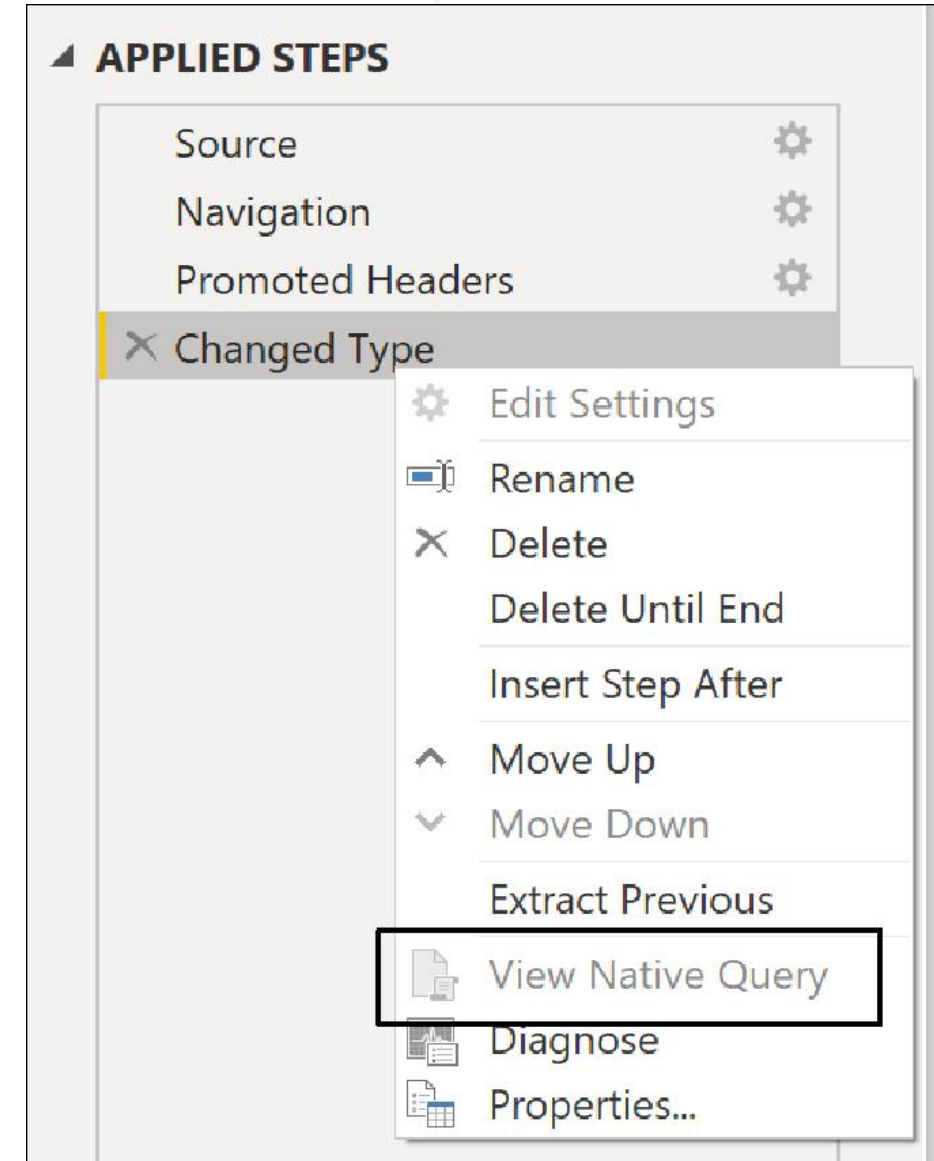


Query folding

Pushes data transformations to the source for better performance and efficiency.

Supported sources

- Relational databases
- OData feeds
- Active Directory





Performance recommendations

- Only keep necessary data
- Check data types
- Reduce cardinality
- Disable query load
- Use parameters

Design a semantic model in Power BI





Data table types

Fact tables are activities or events.

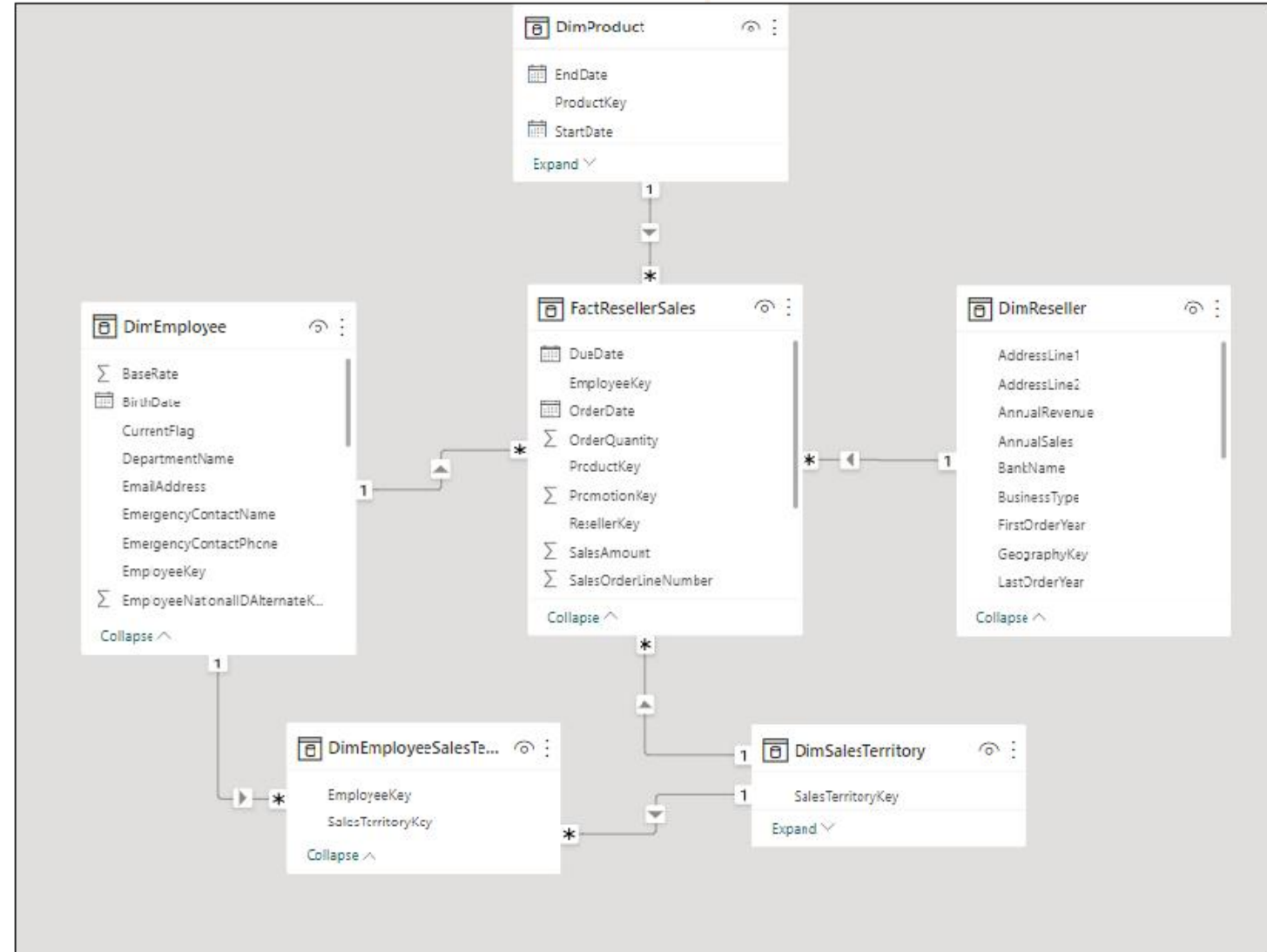
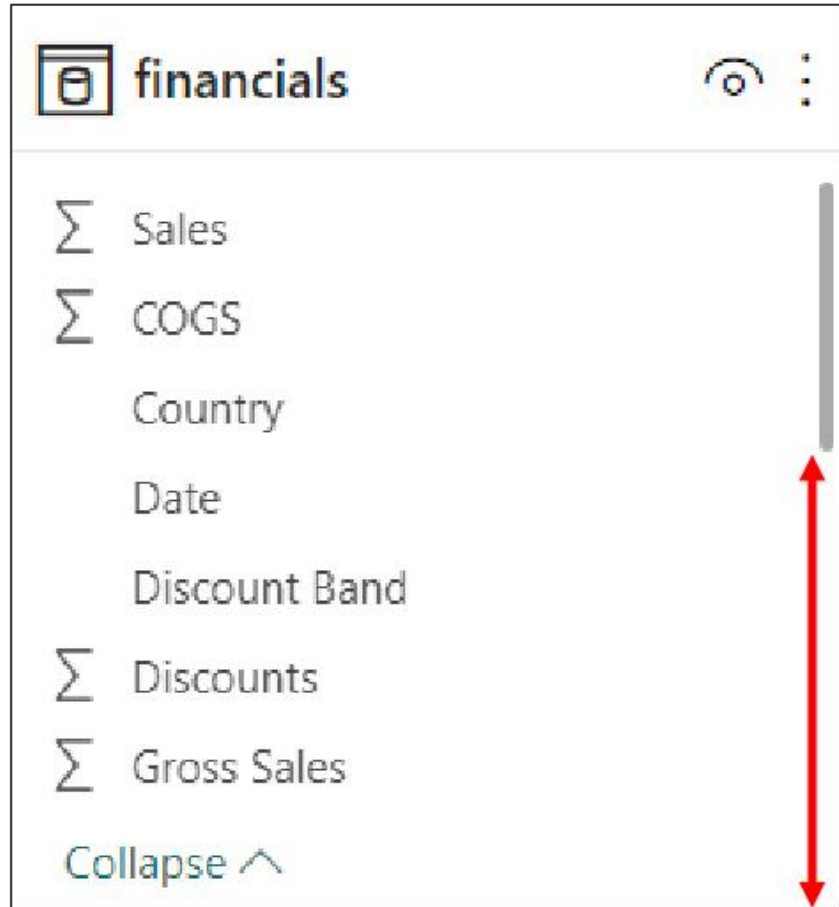
FactResellerSales	
DueDate	
EmployeeKey	
OrderDate	
OrderQuantity	
ProductKey	
PromotionKey	
ResellerKey	
SalesAmount	
SalesOrderLineNumber	
Collapse ^	

Dimension tables provide the details.

DimEmployee	
BaseRate	
BirthDate	
CurrentFlag	
DepartmentName	
EmailAddress	
EmergencyContactName	
EmergencyContactPhone	
EmployeeKey	
EmployeeNationalIDAlternateK...	
Collapse ^	



Understand star schemas





Create relationships

Manage relationships

Active	From: Table (Column)	To: Table (Column)
☒	DimEmployeeSalesTerritory (EmployeeKey)	DimEmployee (EmployeeKey)
☒	DimEmployeeSalesTerritory (SalesTerritoryKey)	DimSalesTerritory (SalesTerritoryKey)
☒	FactResellerSales (EmployeeKey)	DimEmployee (EmployeeKey)
☒	FactResellerSales (ProductKey)	DimProduct (ProductKey)
☒	FactResellerSales (ResellerKey)	DimReseller (ResellerKey)
☒	FactResellerSales (SalesTerritoryKey)	DimSalesTerritory (SalesTerritoryKey)

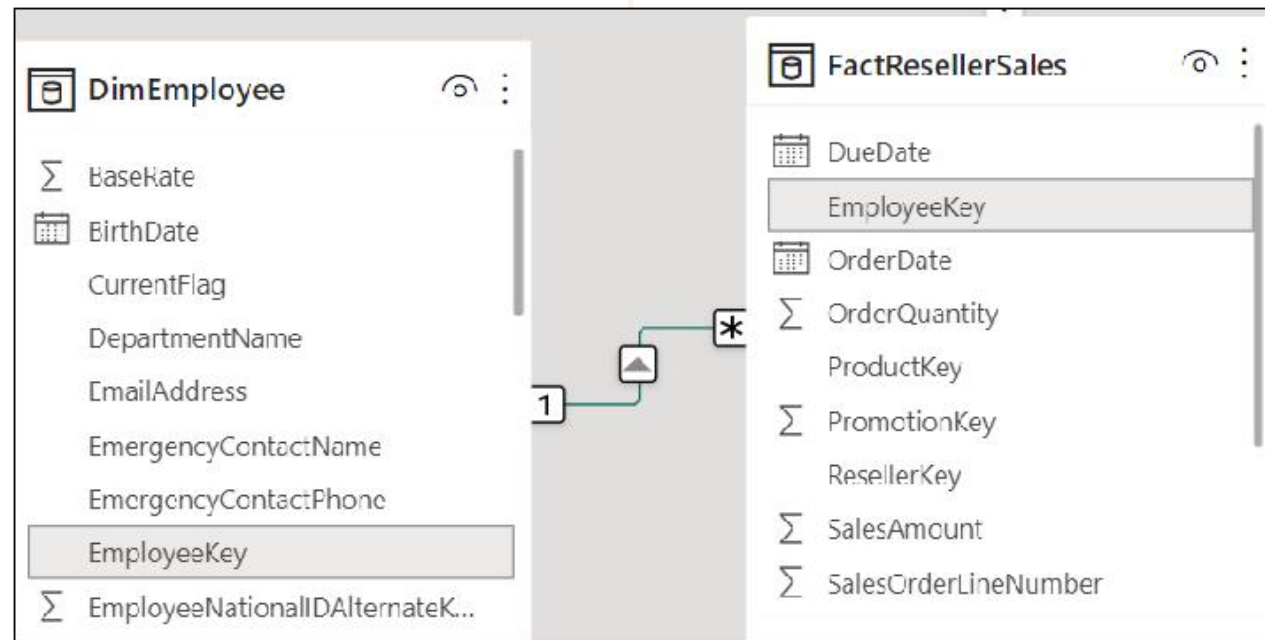
New...

Autodetect...

Edit...

Delete

Close





Edit relationships

Edit relationship

Select tables and columns that are related.

FactResellerSales

DueDate	ShipDate	ProductKey	ResellerKey	PromotionKey	EmployeeKey	SalesTerritoryKey
Monday, September 4, 2017	Friday, September 1, 2017	235	312	1	282	
Monday, September 4, 2017	Friday, September 1, 2017	351	312	1	282	
Monday, September 4, 2017	Friday, September 1, 2017	348	312	1	282	

DimEmployee

EmployeeKey	ParentEmployeeKey	EmployeeNationalIDAlternateKey	ParentEmployeeNationalIDAlternateKey
12	189	912265825	33237992
17	189	132674823	33237992
24	201	835460180	332349500

Cardinality: Many to one (*:1)

Cross filter direction: Single

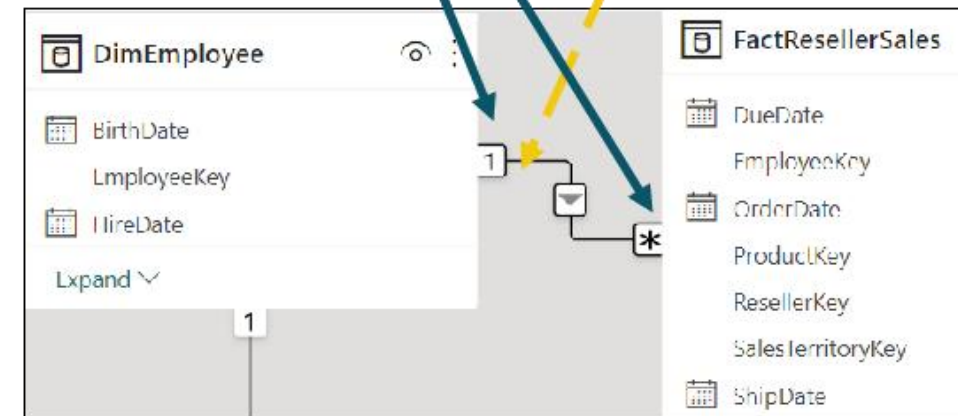
- ☒ Make this relationship active
- ☐ Apply security filter in both directions
- ☐ Assume referential integrity

Cardinality

- Many to one (*:1)
- Many to one (*:1)
- One to one (1:1)
- One to many (1:*)
- Many to many (*:*)

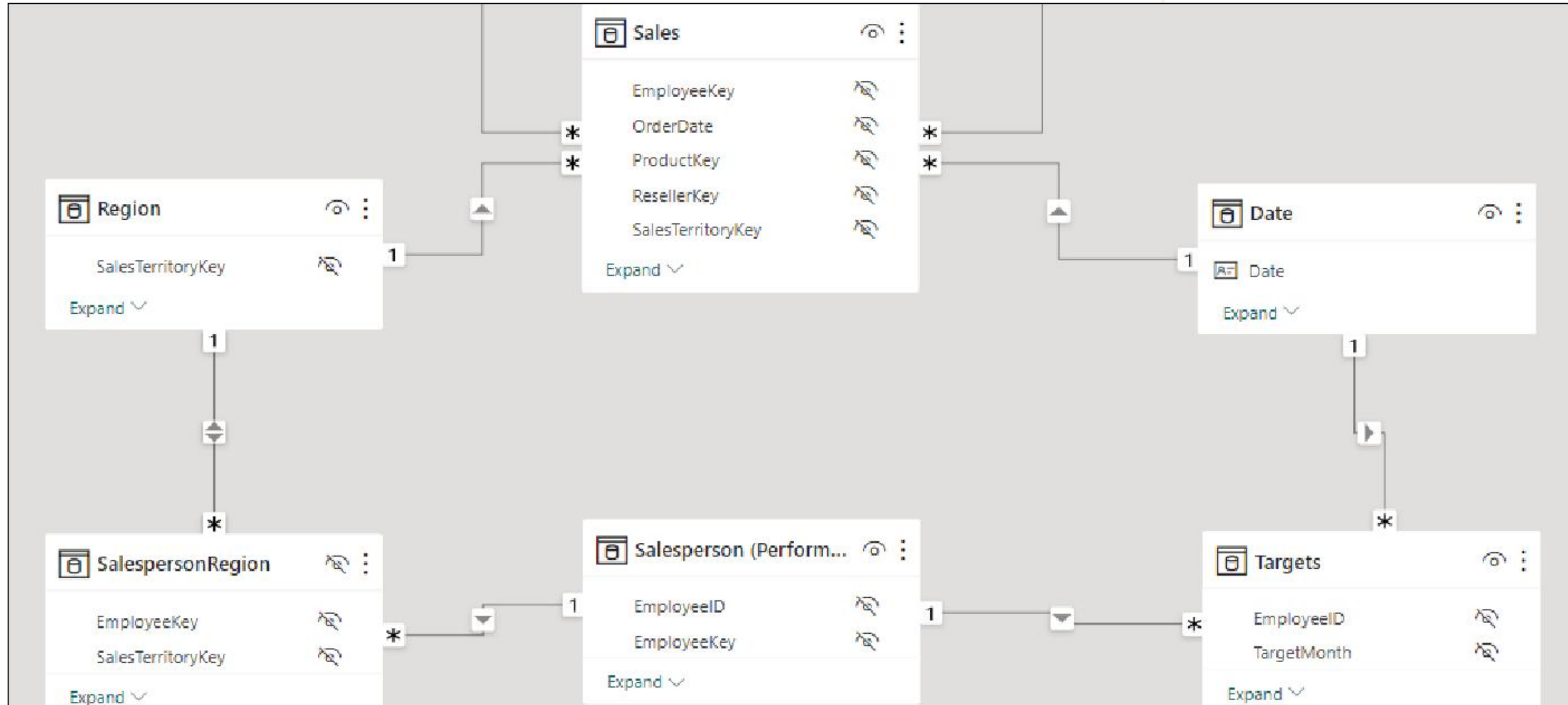
Cross filter direction

- Single
- Single
- Both



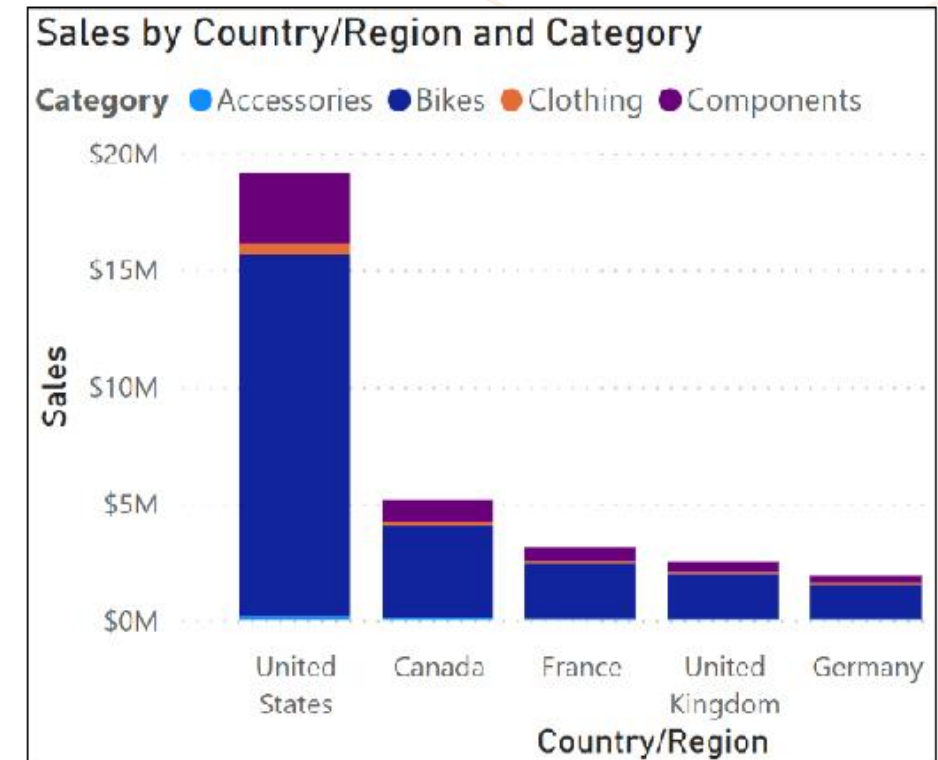
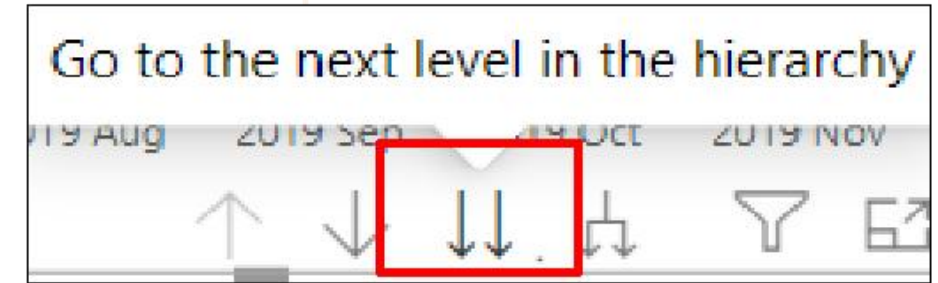
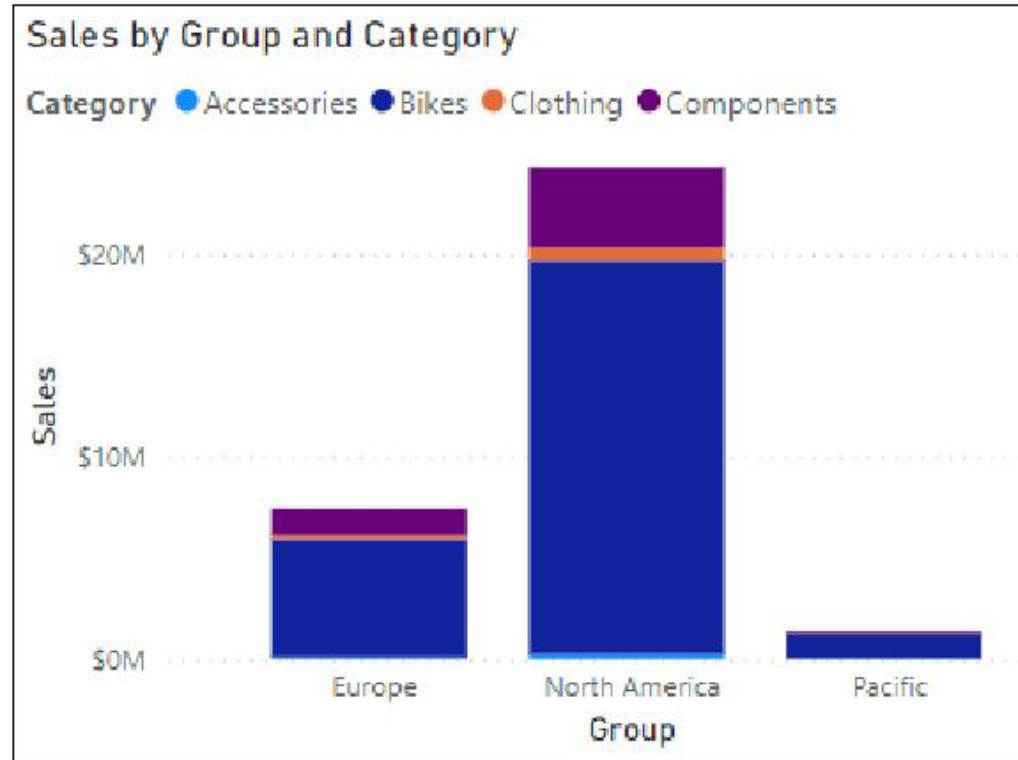
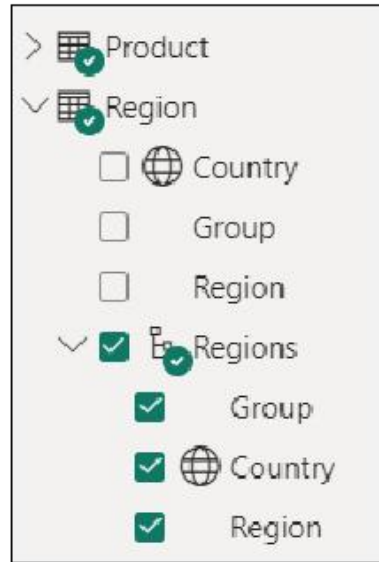


Implications of circular relationships





How to use hierarchies for data fields



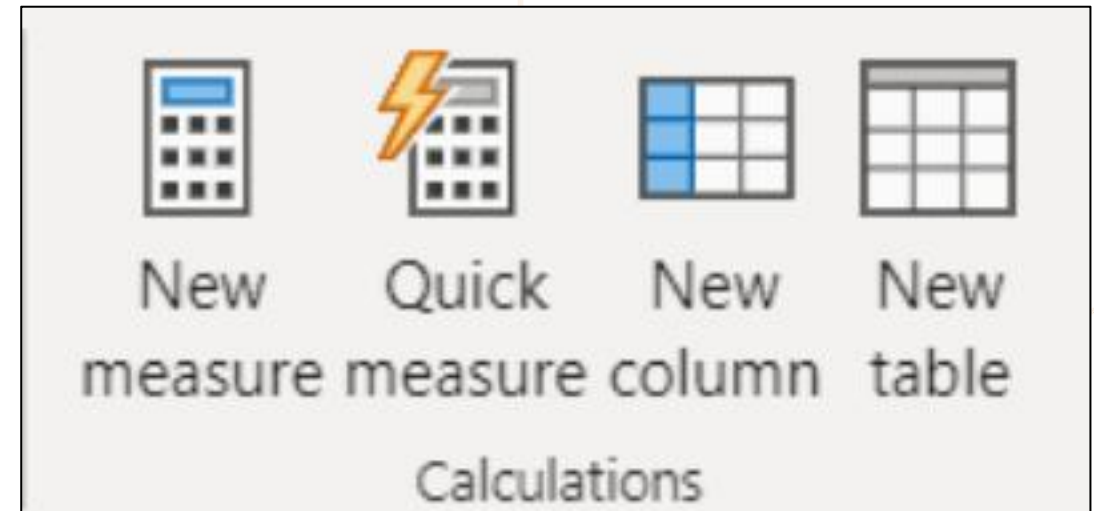
Add measures to semantic
models





What is DAX?

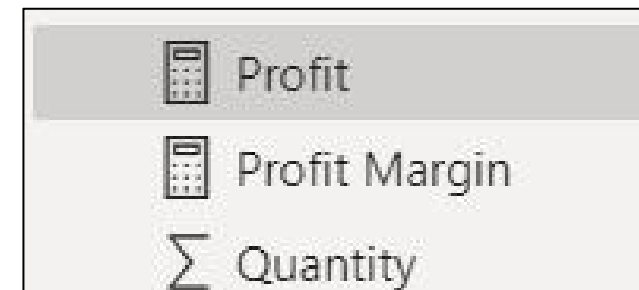
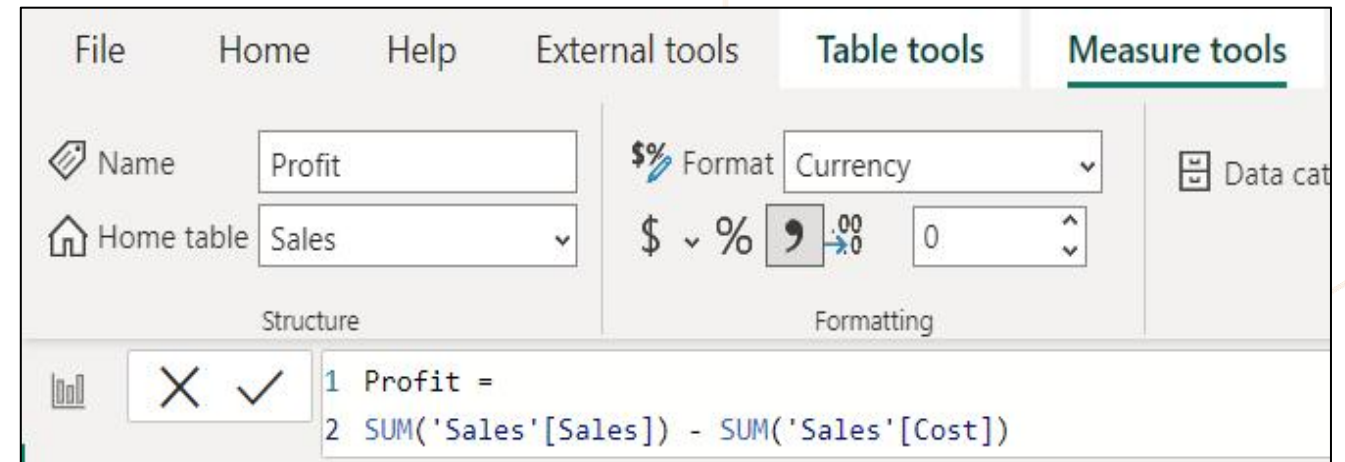
- Data Analysis Expressions
- Library of functions and operators
- Build formulas and expressions
- Create calculated measures, columns, and tables





Create calculated measures

- Defined with DAX definitions
- Computed on the fly.
- Not stored in data model.
- Responsive to interactions.
- Indicated by calculator icon.





Implicit vs. Explicit vs. Quick measures

- ☐  Sales % All Region
- ☐  Sales % Country
- ☐  Sales % Group
- ☐ Σ Sales
- ☐ ShipDate

Quick measure

Calculation

Filtered value 

Calculate a value with a filter applied. [Learn more](#)

Base value ⓘ

Add data fields here

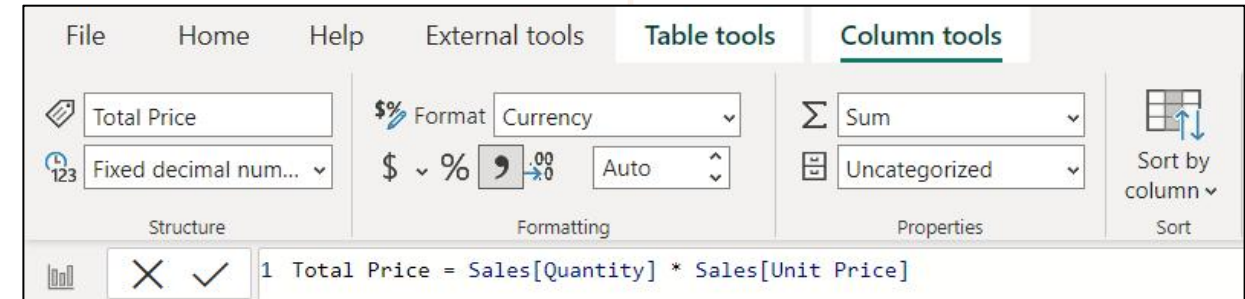
Filter ⓘ

Add data fields here



Create calculated columns

- Defined using DAX expressions.
- Computed & stored in data model.
- Useful “helper/connector columns.”
- Recalculated during data refresh.
- Table and Sigma icon.



Total Price
\$57.68
\$4,049.98
\$4,049.98
\$57.68
\$1,637.4
\$40.38
\$4,049.98



Columns vs. measures

- Calculated columns:
 - Create values for each row in table.
 - Store values in the .pbix file.
 - Increases data model size.
 - Row-by-row calculation can impact performance.
 - Must be referenced with measures for reuse.
- Measures:
 - Calculate on demand.
 - Calculated based on filters.
 - Doesn't affect data model size.
 - DAX expressions may still be suboptimal.
 - Can reference other measures directly for reuse.



Create calculated tables

- Defined using DAX expressions.
- Computed & stored in data model.
- Useful for aggregating data or creating custom tables.
- Table and calculator icon.

The screenshot shows the Power BI Desktop interface with the 'Table tools' ribbon selected. The 'Name' field is set to 'Date'. The 'Structure' section shows the formula bar with the DAX expression: `Date = CALENDAR(AUTO(6))`. The 'Calculations' section shows icons for 'New measure', 'Quick measure', 'New column', and 'New table'. The 'Data' pane on the right shows the 'Date' table selected, with a search bar and a list of columns: Date, Fiscal, Month, MonthKey, Quarter, and Year.

Date	Year	Quarter	Month	MonthKey
7/1/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/2/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/3/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/4/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/5/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/6/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/7/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/8/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/9/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/10/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707



Create a common date table

Mark as date table

Select a column to be used for the date. The column must be of the data type 'date' and must contain only unique values. [Learn more](#)

Date column

Date

✓ Validated successfully

When you mark this as a date table, the built-in date tables that were associated with this table are removed. Visuals or DAX expressions referring to them may break. [Learn how to fix visuals and DAX expressions](#)

OKCancel

✓ Date

☐ Date

✓ ☐ Fiscal

☐ Year

☐ Quarter

☐ Month

☐ Month

☐ Quarter

☐ Year

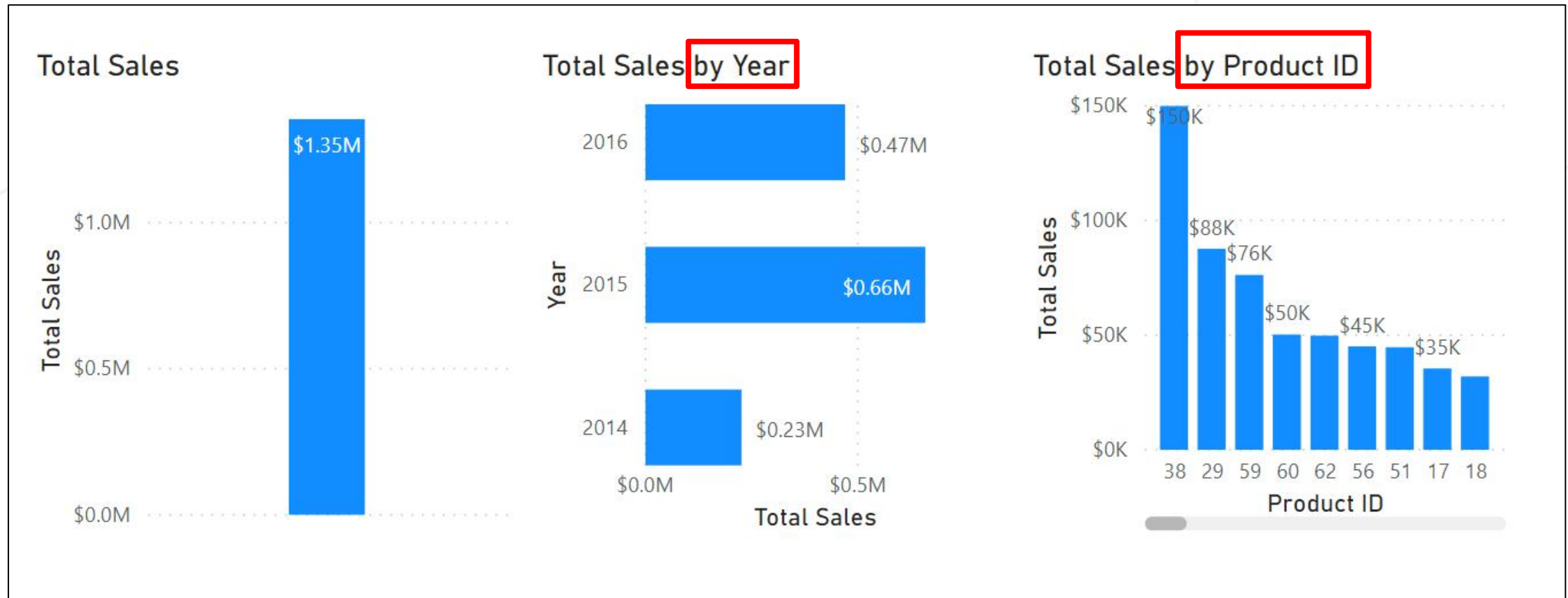
Use DAX functions





Understanding filter context

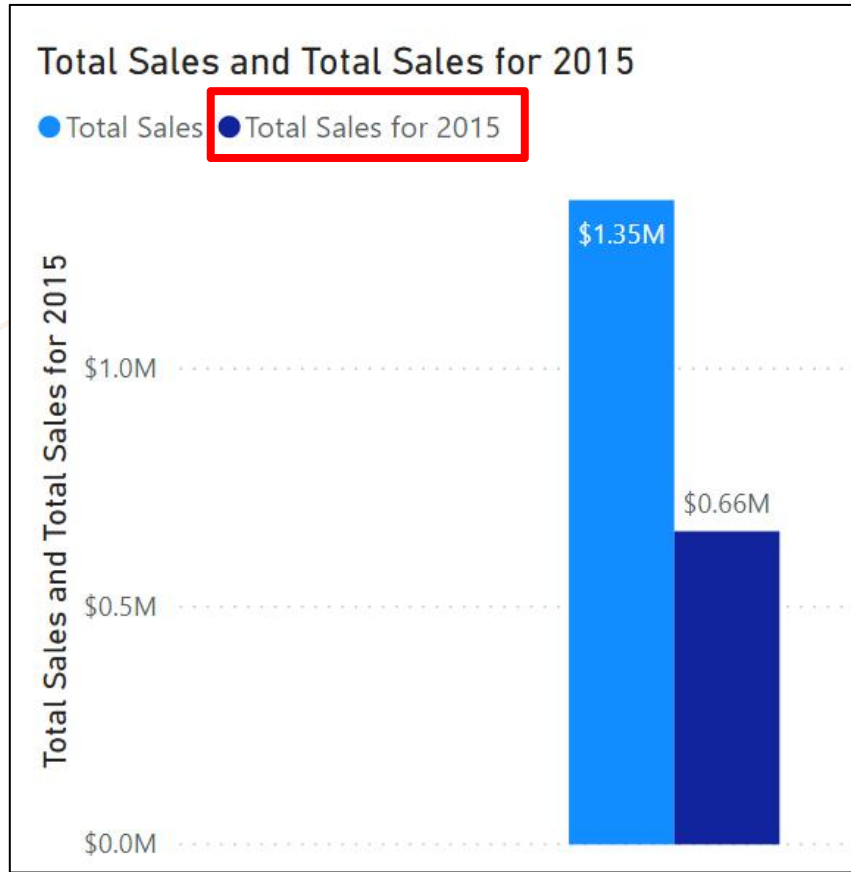
- Measures are contextually different, or “dynamic,” depending on filters.





CALCULATE() function

- Adjusts how measures interpret data filters, enabling context control.



```
Total Sales for 2015 =  
CALCULATE (  
    SUM ( 'Sales OrderDetails'[Total  
Price] ),  
    YEAR ( 'Sales OrderDetails'[OrderDate] )  
= 2015  
)
```

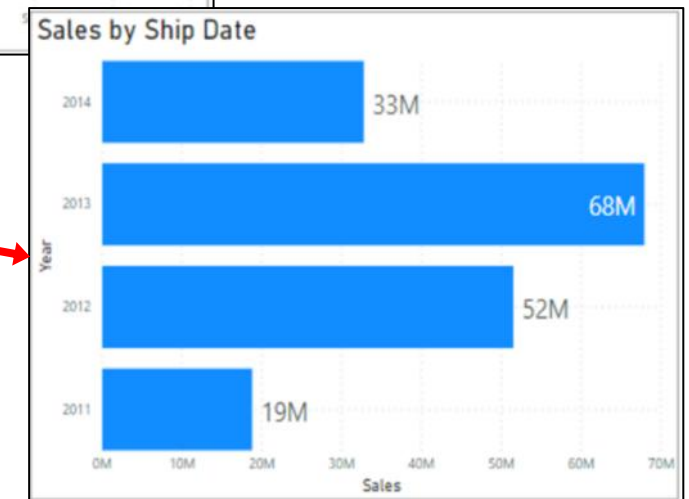


Use inactive relationships with DAX

- Enable additional table filtering without impacting the active relationship.



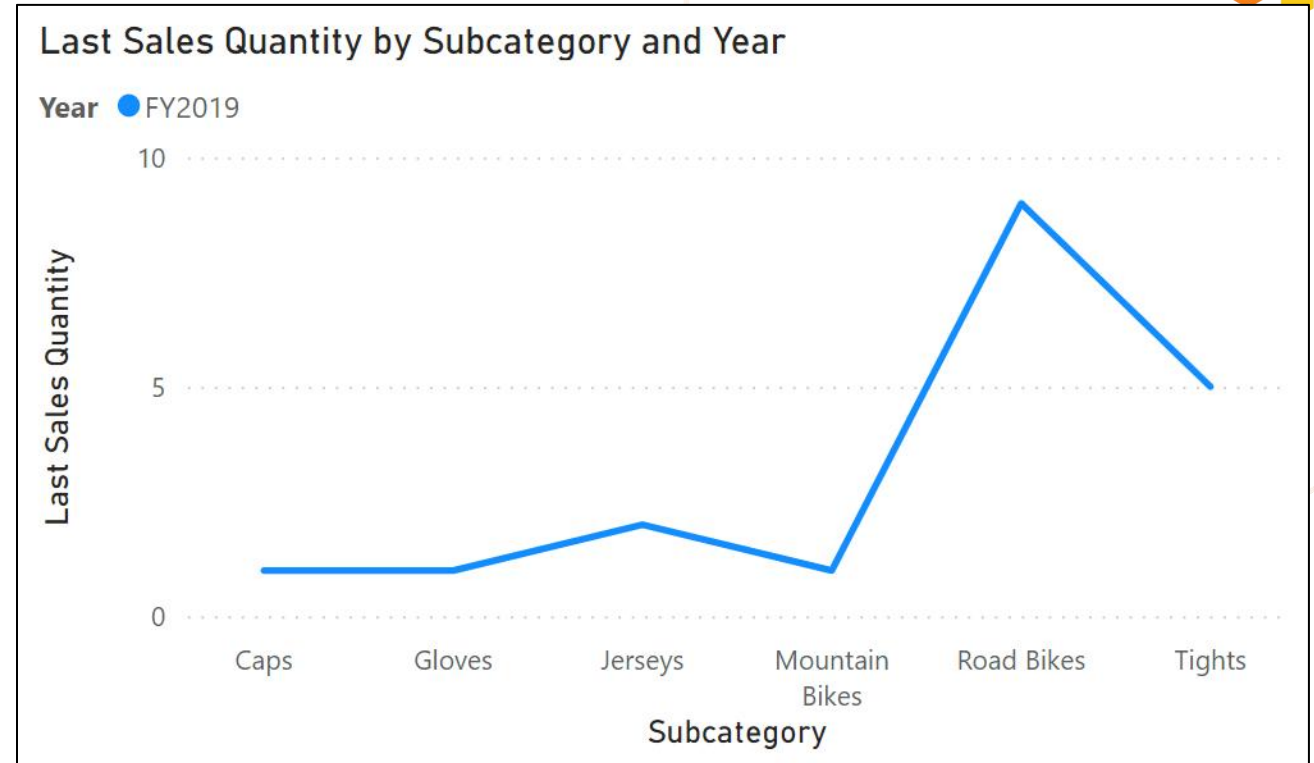
```
Sales by Ship Date =  
CALCULATE (  
    Sales[TotalPrice],  
    USERELATIONSHIP (  
        'Calendar'[Date], Sales[ShipDate])  
)
```





Semi-additive measures

- **Context-dependent:**
Results vary based on dimensions.
- **Time Intelligence:**
Useful for averages, ratios, balances.
- **Caution with aggregation:**
Beware of misleading results when aggregating.



```
Last Sales Quantity =  
CALCULATE(  
    SUM(Sales[Quantity]),  
    LASTDATE('Date'[Date])  
)
```



Create a common date table

Mark as date table

Select a column to be used for the date. The column must be of the data type 'date' and must contain only unique values. [Learn more](#)

Date column

Date

✓ Validated successfully

When you mark this as a date table, the built-in date tables that were associated with this table are removed. Visuals or DAX expressions referring to them may break. [Learn how to fix visuals and DAX expressions](#)

OKCancel

✓ Date

☐ Date

✓ ☐ Fiscal

☐ Year

☐ Quarter

☐ Month

☐ Month

☐ Quarter

☐ Year



Time intelligence functions

Month	2014	2015	2016
January		\$66,692.8	\$100,854.72
February		\$107,900	\$205,416.67
March		\$147,879.9	\$315,242.12
April		\$203,579.29	\$449,872.68
May		\$260,402.99	\$469,771.34
June		\$299,490.99	\$469,771.34
July	\$30,192.1	\$354,955.92	\$469,771.34
August	\$56,801.5	\$404,937.61	\$469,771.34
September	\$84,437.5	\$464,670.63	\$469,771.34
October	\$125,641.1	\$534,999.13	\$469,771.34
November	\$175,345.1	\$580,912.49	\$469,771.34
December	\$226,298.5	\$658,388.75	\$469,771.34
Total	\$226,298.5	\$658,388.75	\$469,771.34

Total Sales Previous Month =

```
CALCULATE ( SUM ( Sales[Total Price] ),  
PREVIOUSMONTH ( 'Date'[Date] ) )
```

Year	Month	Total Sales	Total Sales Previous Month
2015	March	\$39,979.9	\$41,207.2
2015	April	\$55,699.39	\$39,979.9
2015	May	\$56,823.7	\$55,699.39
2015	June	\$39,088	\$56,823.7
2015	July	\$55,464.93	\$39,088
2015	August	\$49,981.69	\$55,464.93
2015	September	\$59,733.02	\$49,981.69
2015	October	\$70,328.5	\$59,733.02
2015	November	\$45,913.36	\$70,328.5
2015	December	\$77,476.26	\$45,913.36



Use variables to improve performance and readability

Without variable:

Sales YoY Growth =

```
DIVIDE (  
    ( [Sales] - CALCULATE ( [Sales], PARALLELPERIOD ( 'Date'[Date], -12, MONTH ) ) ),  
    CALCULATE ( [Sales], PARALLELPERIOD ( 'Date'[Date], -12, MONTH ) ) )
```

With variable:

Sales YoY Growth =

```
VAR SalesPriorYear = CALCULATE ( [Sales], PARALLELPERIOD ( 'Date'[Date], -12, MONTH ) )  
VAR SalesVariance = DIVIDE ( ( [Sales] - SalesPriorYear ), SalesPriorYear )  
  
RETURN  
SalesVariance
```



Thank you
Let's discuss our collaborations

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